5G Call Flow (RAN) and Channels

Nodes

3 Main Components:
UE (User Equipment)
gNB (next-generation NodeB)
5GC (5G Core)
Inside the 5G Core:
AMF (Access and Mobility Management Function)
UPF (User Plane Function)
SMF (Session Management Function)

The Two "Planes"	
Control Plane (CP)	These are the "Signaling" messages. They are the instructions, the handshakes, and the "paperwork" required to set up a connection. (Path: UE <-> gNB <-> AMF)
User Plane (UP)	User Plane (UP): This is your actual data. Your WhatsApp messages, YouTube packets (Path: UE <-> gNB <-> UPF <-> Internet)

The RRC States	
RRC_IDLE	The phone is "sleeping". The network doesn't know exactly which tower you are at, only which "Area" you are in
RRC_CONNECTED	The phone is "talking." The tower knows exactly who you are, and you have a dedicated radio link
RRC_INACTIVE (New in 5G)	The "suspended" state. The phone sleeps to save battery, but the tower remembers the phone's configuration. This allows the phone to wake up and start sending data

5G Interface

Interface Name	Connects	Purpose
Uu	UE <-> gNB	The Air Interface (Radio)
N1	UE <-> AMF	The NAS (Core) Signaling path
N2	GNB <-> AMF	The Control Plane path to the Core
N3	GNB <-> UPF	The User Data path to the Core
Xn	gNB <-> gNB	Tower-to-Tower communication (for Handovers)

Protocol Stack

The Control Plane Stack (used for RRC messages)	
Layer	**The "Job" in Signaling**
NAS (Non-Access Stratum)	The "Secret Letter" to the Core (AMF). Handles Authentication and Registration
RRC (Radio Resource Control)	The Boss. Manages the radio connection, handovers, and system info.
PDCP (Packet Data Convergence)	The Security Guard. Handles encryption and integrity protection
RLC (Radio Link Control)	The Editor. Fixes errors and breaks big messages into small pieces
MAC (Medium Access Control)	The Scheduler. Decides when the phone can talk and handles the RACH
PHY (Physical Layer)	The Radio. Converts bits into 1s and 0s on the radio waves

The User Plane Stack (actual internet traffic lives)	
Layer	**The "Job" in Signaling**
SDAP (Service Data Adaptation Protocol)	It maps your data to specific Quality of Service (QoS) flows. It's the reason why your voice call stays clear even if your background download slows down
PDCP (Packet Data Convergence)	The Security Guard. Handles encryption and integrity protection
RLC (Radio Link Control)	The Editor. Fixes errors and breaks big messages into small pieces
MAC (Medium Access Control)	The Scheduler. Decides when the phone can talk and handles the RACH
PHY (Physical Layer)	The Radio. Converts bits into 1s and 0s on the radio waves

Layer Failure	
Layer	**If Failure**
NAS (Non-Access Stratum)	The tower is fine, but the Core Network rejected your SIM card (e.g., "Illegal UE")
SDAP (Service Data Adaptation Protocol)	QoS Flow Mapping Mismatch
PDCP (Packet Data Convergence)	Integrity Check Failure
RRC (Radio Resource Control)	The phone and tower couldn't agree on a configuration
RLC (Radio Link Control)	RLC Max Retransmissions
MAC (Medium Access Control)	The "Random Access" (RACH) failed. The phone knocked, but the tower didn't hear it
PHY (Physical Layer)	You have a "Radio Link Failure" (the signal was too weak)

The Bearer (In 5G, we don't just send data blindly. we use Bearers (virtual pipes))	
Signalling	
SRB (Signaling Radio Bearer)	**Used for RRC/NAS messages**
SRB0	For common messages (CCCH)
SRB1	For high-priority RRC messages
SRB2	For lower-priority NAS messages
Data	
DRB (Data Radio Bearer)	**Used for your actual internet data**

The Initial Access Flow

The Initial Access Flow	
Sub-Phases	**Main Job**
DL Synchronisation	Listening
RACH/RRC Procedure	Talking

2. The 4-Step RACH (The "Knock") (MAC)	
Msg1 (Random Access Preamble)	The UE sends a "Preamble" on the PRACH channel. There is no identity here; it's just a signal saying "I am here!"
Msg2 (Random Access Response – RAR)	The gNB hears the knock and responds. It provides a Temporary C-RNTI (an ID) and a Timing Advance (telling the phone exactly when to transmit so the signal arrives at the right time)

1. Step 0: The "Search" (Cell Search & MIB/SIB)	
PSS/SSS (Primary/Secondary Sync Signals)	The UE scans frequencies to find these. They provide the Physical Cell ID (PCI) and time/frequency alignment
MIB (Master Information Block)	Sent on the PBCH channel. It gives the UE the "minimum essentials" to find the next message
SIB1 (System Information Block 1)	This is the most important broadcast. It contains the PLMN ID (Operator name), cell barring info, and the RACH Configuration (the instructions on how to "knock")

3. The L3 Signaling (The "Handshake") (RRC)		
Msg3: RRCSetupRequest	UE -> gNB	The UE identifies itself and says why it's calling (e.g., emergency, mo-data)
Msg4: RRCSetup	GNB -> UE	The tower sends the "Radio Rules." It configures SRB1 (the high-priority signaling pipe)
Msg5: RRCSetupComplete	UE -> gNB	The UE says "Got it." Inside this message, it hides a NAS Registration Request for the Core

4. Security & Registration (The "ID Check")	
Authentication	The Core challenges the SIM card to prove it's valid
Security Mode Command	The gNB and UE turn on Encryption and Integrity Protection (the PDCP layer)
RRCReconfiguration	This is the "Big Boss" message. It sets up the DRBs (Data Radio Bearers) and the SDAP layer so you can finally send internet data

RRC Message

From – To	Log Message Name	Inside message
UE – gNB	PRACH Preamble	Index: 42
GNB – UE	MAC Random Access Response	Timing Advance: 12, Temp C-RNTI: 0x1A2B
UE – gNB	RRCSetupRequest	Cause: mo-Data, ID: 0x554433
GNB – UE	RRCSetup	SRB1 Configured
UE – gNB	RRCSetupComplete	SelectedPLMN: 01, DedicatedNASMessage: 0x0741...
GNB – UE	DLInformationTransfer	NAS: Authentication Request
UE – gNB	ULInformationTransfer	NAS: Authentication Response
GNB – UE	SecurityModeCommand	Integrity: nia2, Ciphering: nea2
UE – gNB	SecurityModeComplete	
GNB – UE	RRCReconfiguration	Contains: radioBearerConfig, secondaryCellGroup
-------------2000 ms GAP-------------		
UE – gNB	RRCReestablishmentRequest	Cause: reconfigurationFailure

Mobility (Handover)

1. The Handover Strategy: "Mobile Assisted"

Measurement Control	The gNB sends an RRCReconfiguration message to the UE, telling it: "Listen to neighboring towers on these frequencies and tell me if they get stronger than me."
Measurement Report	When a neighbor's signal exceeds a threshold (called Event A3), the UE sends a MeasurementReport to the tower

2. The Signaling Flow (The "Execution")

Step	Message	Interface	Description
1	Handover Request	Xn / N2	The Source gNB asks the Target gNB: "Can you take this user?"
2	Handover Request Ack	Xn / N2	Target gNB says "Yes" and sends the new radio configurations.
3	RRCReconfiguration	Uu (Air)	The Handover Command. The Source tower tells the UE: "Switch to the Target tower NOW."
4	RACH Preamble	Uu (Air)	The UE "knocks" on the new Target tower.
5	RRCReconfigurationComplete	Uu (Air)	The UE confirms it has successfully landed on the new tower.

3. Handover Failures

HO Failures	Reason	Result	Fix
A. "Too Late" Handover	The UE is moving too fast. By the time the gNB sends the RRCReconfiguration (Handover Command), the signal from the source tower is already too weak for the UE to decode it.	Radio Link Failure (RLF) followed by an RRC Re-establishment	Slow down
B. "Too Early" Handover (Ping-Pong)	The UE moves to the target tower, but then immediately sees the source tower as stronger and switches back.	High signaling load and "choppy" data throughput	Adjust the Hysteresis or Time-to-Trigger (TTT) parameters in the Measurement Config
C. Target Cell Access Failure	The UE receives the Handover Command and tries to "knock" (RACH) on the target tower, but the target tower is congested or has a hardware fault	The UE tries to go back to the source tower (Handover Failure), or the call drops	Clear the HW fault

4. Key 3GPP Measurement Events

Event A1	Serving cell becomes better than a threshold (Stop searching for neighbors)
Event A2	Serving cell becomes worse than a threshold (Start searching for neighbors)
Event A3	Neighbor becomes offset better than Serving (The classic Handover trigger)
Event A5	Serving becomes worse than threshold1 AND Neighbor becomes better than threshold2 (The "Desperation" handover)

5G Channels

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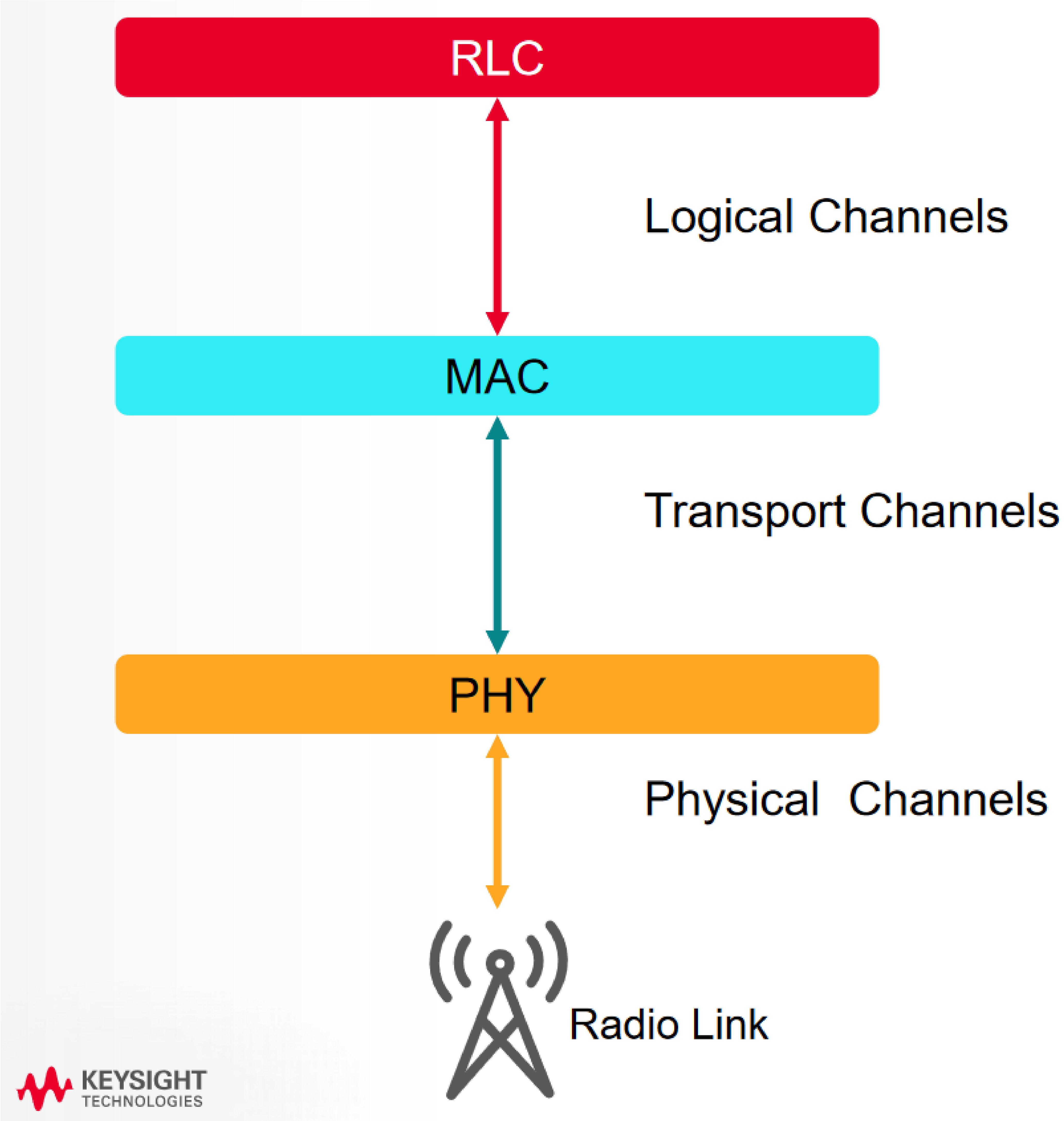
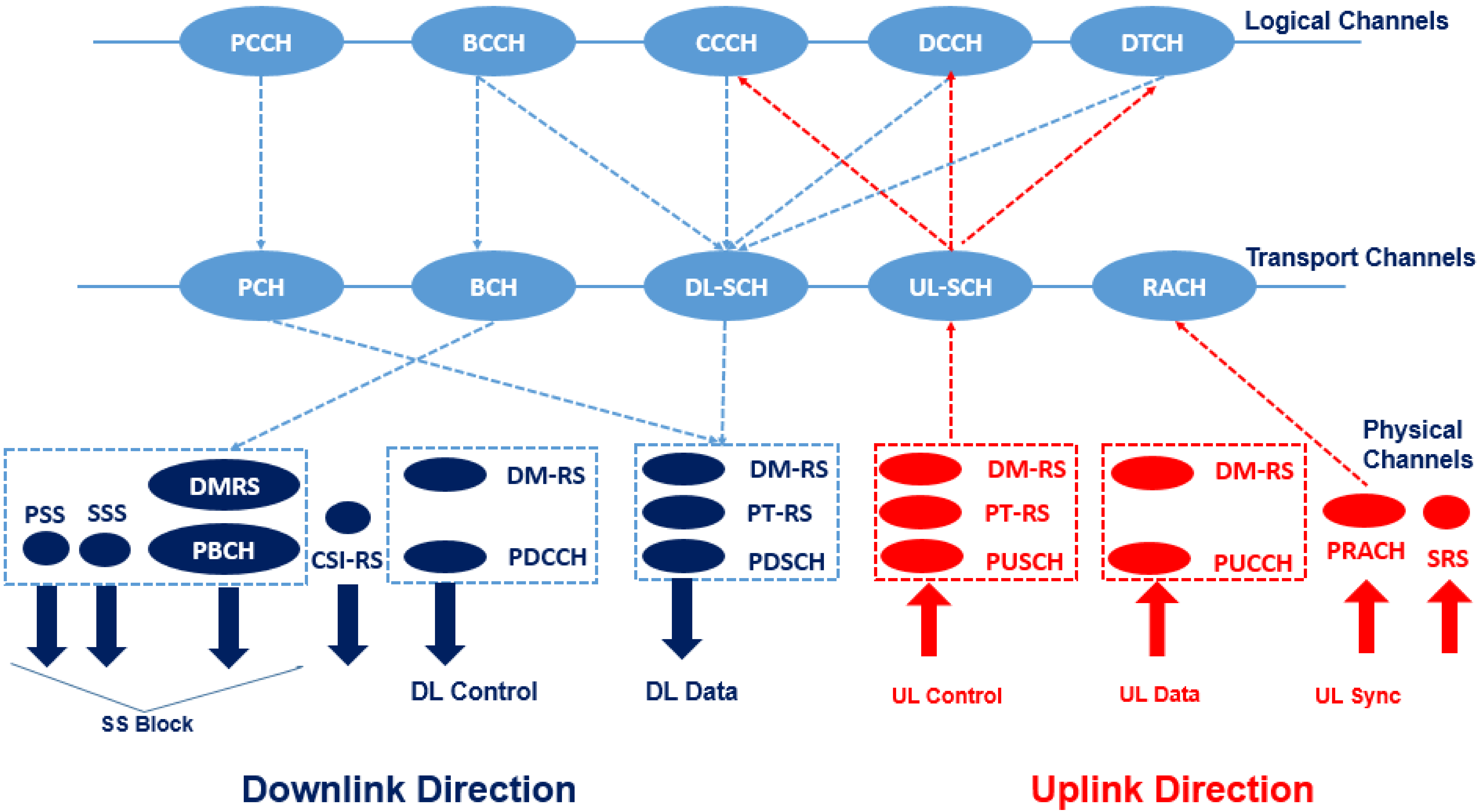


Define **the type of data to be transferred:** traffic (user data), paging messages, dedicated control information…

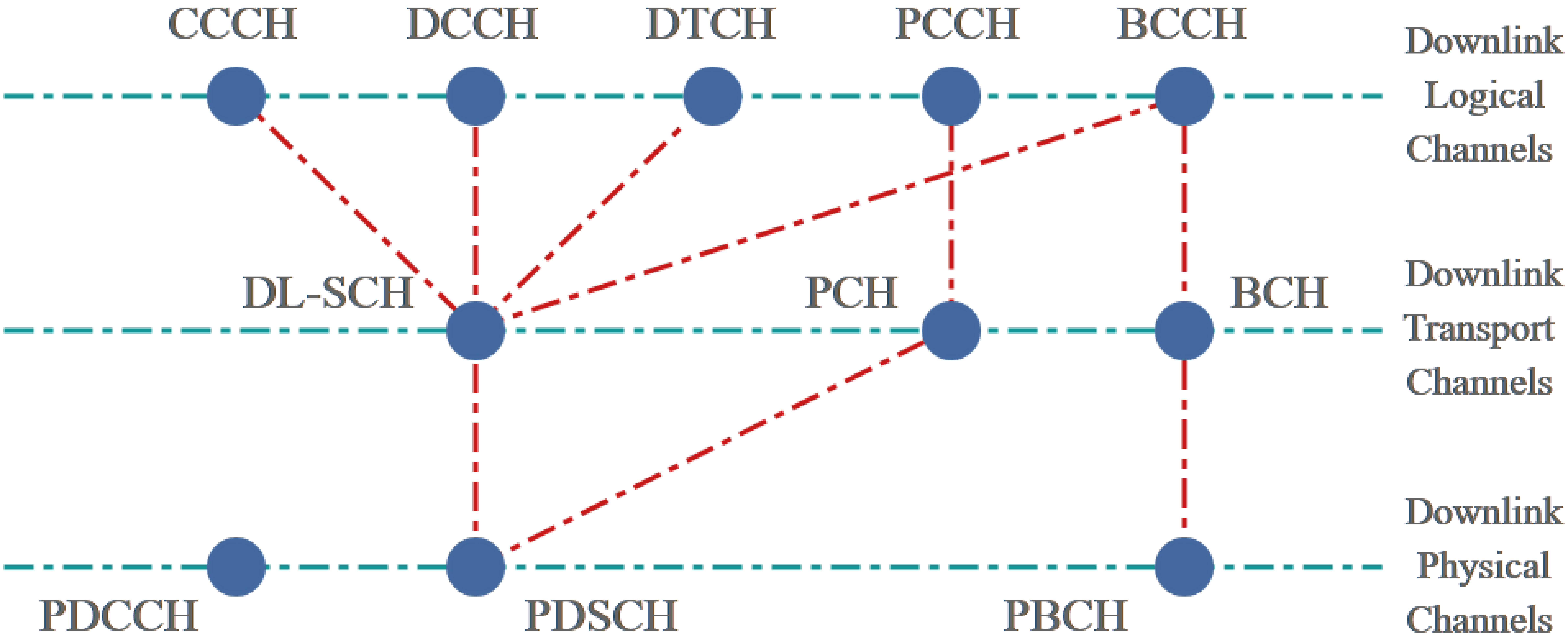
Define **how the information will be carried to the physical layer** and the characteristics of the data: error protection, channel coding and CRC, data packet size…

Characterized by their timing and access protocols (ex: random access channels), **data rates** (ex: traffic channels)…

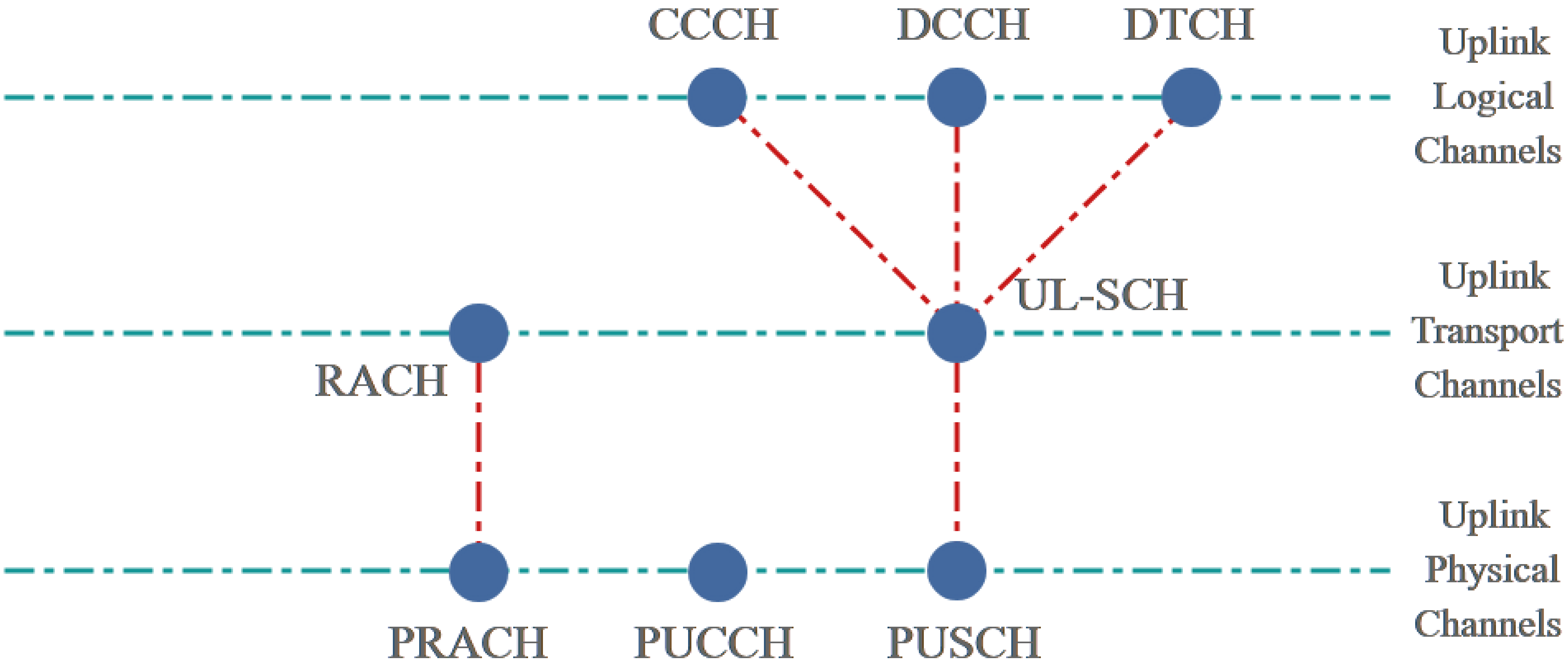
5G Channels


Logical Channels
PCCH
BCCH
CCCH
DCCH
DTCH
Transport Channels
PCH
BCH
DL-SCH
UL-SCH
RACH
Physical Channels
PSS
SSS
DMRS
PBCH
CSI-RS
DM-RS
PDCCH
DM-RS
PT-RS
PDSCH
DM-RS
PT-RS
PUSCH
DM-RS
PUCCH
PRACH
SRS
SS Block
DL Control
DL Data
UL Control
UL Data
UL Sync
Downlink Direction
Uplink Direction

5G Channels - Downlink



5G Channels - Uplink



5G Channels

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Channel Categories		
Channels	**Layer**	**Job**
Logical Channels	MAC & RLC	They tell the network if the content is for control or for the user.
Transport Channels	MAC & PHY	They define the characteristics of how the data will travel over the air.
Physical Channels	PHY	These are the actual radio signals. This is where your PHY Layer (from your checklist) handles modulation and coding.

Logical Channels		
Channel	**Full Name**	**Easy Explanation**
BCCH	Broadcast Control	The "Store Sign." Tells every phone the cell ID and basic rules.
PCCH	Paging Control	The "Pager." Tells a sleeping phone "Hey, you have an incoming call!"
CCCH	Common Control	The "Front Desk." Used by phones that haven't fully connected yet (SRB0).
DCCH	Dedicated Control	The "Private Intercom." For specific radio/core commands (SRB1/2/3).
DTCH	Dedicated Traffic	The "Cargo Belt." Carries your actual internet data (DRBs).

5G Channels

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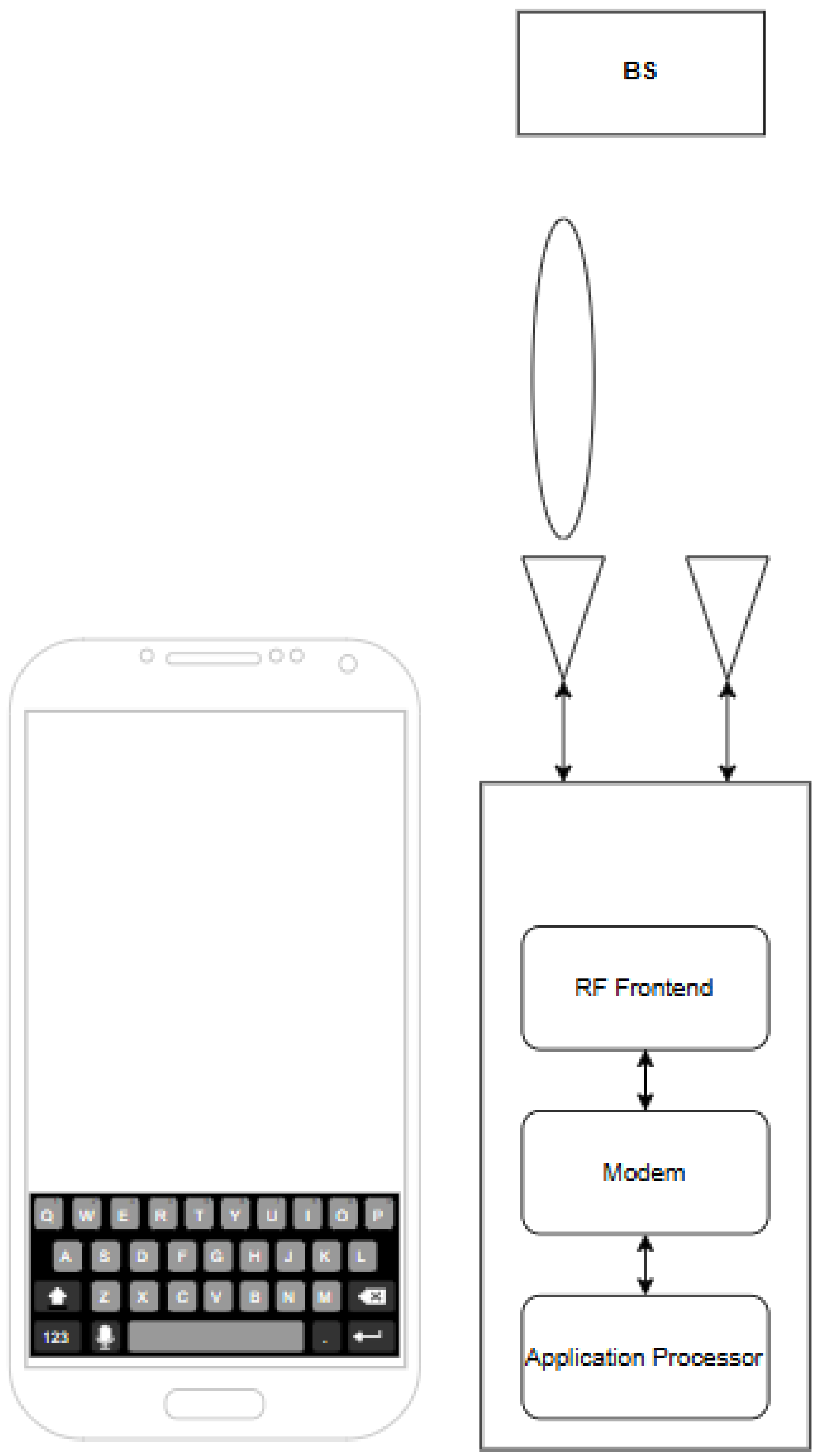
Transport Channels		
Channel	**Full Name**	**Easy Explanation**
BCH	Broadcast	The "Public Radio." Fixed format, very robust, carries the BCCH info.
PCH	Paging	The "Wake-up Signal." Carries the PCCH info to the physical layer.
DL-SCH	Downlink Shared	The "Main Delivery Truck." Carries all user data and most signaling to the phone.
UL-SCH	Uplink Shared	The "Return Truck." Carries all data and signaling from the phone to the gNB.
RACH	Random Access	The "Door Knock." The only way a phone can start a conversation.

Physical Channels		
Channel	**Full Name**	**Easy Explanation**
PBCH (Downlink)	Physical Broadcast	The "Public Radio." Fixed format, very robust, carries the BCCH info.
PDCCH (Downlink)	Physical Downlink Control	The "Wake-up Signal." Carries the PCCH info to the physical layer.
PDSCH (Downlink)	Physical Downlink Shared	The "Main Delivery Truck." Carries all user data and most signaling to the phone.
PRACH (Uplink)	Physical Random Access	The "Return Truck." Carries all data and signaling from the phone to the gNB.
PUCCH (Uplink)	Physical Uplink Control	The "Door Knock." The only way a phone can start a conversation.
PUSCH (Uplink)	Physical Uplink Shared	The "Uplink Bulk." Carries your voice, uploads, and NAS signaling.

Initial UE Attach



1. Press Power On (or Airplane mode off) button on UE(User Equipment, e.g, mobile phone) display
2 & 3 - The Power On signal goes to Application Processor (AP)
4. The signal is translated (recognized) as 'Radio ON' command in AP chip
5 & 6 - The 'Radio On' command is transferred to Modem chip (MODEM)
7. Once the modem chip recieves the 'Power ON' command, it performs a sequence of internal procedure and initiate 'RACH' process
8. The RACH trigger is conveyed to RF Front End Module (FEM)
9. A command from FEM goes to RF hardware (e.g, Antenna)
10 – 14. The RF hardware transmit the signal through the air (channel) and reaches to the RF hardware(antenna) of Base station.
15. The RF signal is detected and processed in the base station.
16-21. The response signal is transmitted from Base Station antenna and conveyed to UE Antenna.
22. The received signal reaches RF front end chip
23. The signal is conveyed to Modem, processed there and recognized as 'RACH Response' signal
24. Now the modem chip generated RRC Request message
25. The generated message is transferred to RF Front End
26. The message goes through RF Front End and relayed to RF hardware (e.g, Antenna)
27-31. The RF hardware transmit the signal through the air (channel) and reaches to the RF hardware(antenna) of Base station
32. The RF signal is detected and processed in the base station
33. Usually during the RRC Request/Setup process, some messages(Registration Request in case of 5G) destined to Core Network is embedded into the RRC Message and that message is relayed to Core Network
34 & 35. The message is processed within the Core Network
36. The response signal is generated by Base State and transmitted into air
37-43. The response signal from Base Station antenna is propogated to UE Antenna.
44. The received signal reaches RF front end chip
45 & 46. The signal is conveyed to Modem, processed there and recognized as 'Response' signal(message)
47. Core Network initiates and generate a sequence of message for Authentication and Security check
48. The message is conveyed to Base Station
49-52. The base station transmit the message into the air and it reaches to UE antenna
53. The message is transferred to RF Front End Chip
54. The message is transferred to Modem Chip
55. The Modem chip process the received message and recognize/decode Authentication and Security check
56. If Modem chip generate the response message for the received message
57. The message/signal is transferred to RF Front End Chio
58. The signal is transferred to RF hardware (antenna)
59-63. The signal is transmitted into air and reaches Base Station
64. The message is received by Base Station and transmitted to Core Network
65. The core network process the received message and this completes the initial registration
66. Now you get the 5G icon with antenna bars on the UE's display

THANK YOU

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